

Chia-Suan Yu

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EDUCATION

Carnegie Mellon University

Aug. 2026 – Dec. 2027 (Expected)

M.S. in Computer Vision, School of Computer Science

Pittsburgh, PA

- Planned courses: *Advanced Computer Vision, Deep Learning System, Introduction to Robot Learning*

National Tsing Hua University

Sep. 2021 – Jun. 2025

B.S. in Computer Science

Hsinchu, Taiwan

- Selected courses: *Introduction to Machine Learning, Robotic Navigation, Design and Analysis of Algorithms, Linear Algebra, Probability, Operating Systems, Computer Architecture*

RESEARCH & WORK EXPERIENCE

Institute of Information Science, Academia Sinica

Mar. 2026 – Jul. 2026

Research Assistant, IISCVML Lab (Supervised by Dr. Tyng-Luh Liu)

Taipei, Taiwan

FlowGuard: Success-Only Runtime Monitoring for Receding-Horizon Generative Robot Policies
(Under Submission to **NeurIPS 2026**)

- Co-developed a success-only failure-detection layer for receding-horizon generative robot policies to safeguard closed-loop robotic manipulation.
- Engineered a dual-stage monitoring pipeline using **VQ action tokenizers** for pre-execution chunk screening and **consistency models** for post-execution failure detection.
- Evaluated the framework on the **RoboMimic** benchmark, achieving a macro **AUROC of 0.912** while controlling the false positive rate under 6%.

VLNVerse Challenge – ECCV 2026 Workshop on Embodied Multimodal Reasoning (EMR)

- Developed physically-grounded embodied agents, designing multimodal architectures to optimize multi-task navigation and long-horizon spatio-temporal reasoning.

Microsoft

Mar. 2025 – Sep. 2025

Research Development Intern, FY2025 (Customer Success Unit)

Taipei, Taiwan

- Built intelligent itinerary planning and generative AI systems using the **Semantic Kernel** multi-agent framework and Azure AI Foundry for real-world enterprise use cases.
- Utilized Azure Function App and MCP Tools to package and deploy customized AI agents to Azure Cloud, enabling real-time access via **Server-Sent Events (SSE)**.
- Designed an automated PII text masking workflow with Azure AI PII Detection API.
- Researched emerging web technologies NLWeb and developed technical prototypes supporting the supervisor's presentation at **Microsoft DevDays Asia 2025**.

National Tsing Hua University

Jan. 2024 – Jul. 2025

Research Assistant, HMI Lab (Supervised by Dr. Po-Chih Kuo)

Hsinchu, Taiwan

A Deep Learning Model Based on Ultrasound Imaging to Differentiate Malignant from Benign Pleural Effusion: a multicenter cohort study (Published in *Respiratory Research*, 2026)

In partnership with National Taiwan University Hospital

- Engineered a deep learning classification model based on **ResNet-18** to non-invasively identify malignant pleural effusion (MPE) using thoracic ultrasound images.
- Trained and fine-tuned the model on **1,706 clinical samples** across multicenter cohorts to ensure robust generalization, achieving a peak **AUC of 0.81**.
- Served as the First Author and Presenter, winning the **Outstanding Academic Paper Award** (Oral Presentation) at the 2024 Annual Congress of the TSPCCM.

- Designed and supervised coding assignments for over 200 students and guided engineering teams in completing their deep learning final projects.

PUBLICATIONS

- [1] Chang-Wei Wu, **Chia-Suan Yu**, Yen-Lin Chen, Po-Chih Kuo, Meng-Rui Lee, Jann-Yuan Wang, Chao-Chi Ho, Jin-Yuan Shih, & Hao-Chien Wang. (2026). **A deep learning model based on ultrasound imaging to differentiate malignant from benign pleural effusion: a multicenter cohort study.** *Respiratory Research*. <https://doi.org/10.1186/s12931-026-03574-w>
- [2] Pei-Chi Chen, **Chia-Suan Yu**, Tsung-Wei Ke, Hwann-Tzong Chen, Tyng-Luh Liu. **FlowGuard: Success-Only Runtime Monitoring for Receding-Horizon Generative Robot Policies.** Under Submission to *NeurIPS 2026*.

HONORS & AWARDS

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| 2024 | Outstanding Academic Paper Award (Oral Presentation) , Annual Congress of the Taiwan Society of Pulmonary and Critical Care Medicine (TSPCCM) |
| 2024 | 2nd Prize , Mobile Communications Practice Competition |

PROJECTS

- Pikachu Seeker: Autonomous Robot Navigation** |  **Repo** Jun. 2025
- Integrated an autonomous robot in a **Unity** simulator using **ROS 2** and **YOLO** object detection for target tracking under RGB-only conditions.
- Tetris Battle AI Agent** |  **Repo** Oct. 2023 – Jan. 2024
- Implemented a **Deep Q-Network (DQN)** reinforcement learning model with experience replay and adaptive reward functions.
 - Showed superhuman performance by consistently defeating human players in real-time gameplay evaluation.
- Smart Energy Saving IoT System** |  **Repo** Oct. 2024 – Dec. 2024
- Developed an automated environmental monitoring system using **ESP32** and **MQTT** protocol for real-time data transmission.
 - **Awarded 2nd Prize** in the 2024 Mobile Communications Practice Competition.
- BOYO Welfare Foundation Rental Platform** |  **Repo** Mar. 2024 – Jun. 2024
- Developed a full-stack rental platform for an early intervention organization using **Laravel (PHP)** with database integration.
- Wallistic: Mobile Wallpaper App** |  **Repo** Feb. 2023 – Jul. 2023
- Designed and deployed a cross-platform mobile application using **React Native** to optimize user time and space efficiency.

SKILLS

Programming:	Python, C/C++, Verilog, JavaScript, PHP, HTML/CSS
Frameworks & Tools:	Git, React Native, ReactJS, Laravel, TensorFlow, PyTorch, OpenCV, Azure, Docker
Expertise:	Machine Learning, Computer Vision, Large Language Models, Web Development
Languages:	Mandarin (Native), English (Proficient)